FlowGRN: Latent Flow Matching for Gene Regulatory Network Inference

A Detailed Research Proposal

---

1 Architecture Design

1.1 Background: The DeepSEM/DAZZLE Paradigm

The current leading deep learning approach on BEELINE is based on the **Structural Equation Model (SEM)** framework, used by DeepSEM and its successor DAZZLE.

Linear SEM assumption:

$$\boldsymbol{X} = \boldsymbol{X}\boldsymbol{A}^\top + \boldsymbol{Z}$$

where $\boldsymbol{X} \in \mathbb{R}^{c \times g}$ is the gene expression matrix (c cells, g genes), $\boldsymbol{A} \in \mathbb{R}^{g \times g}$ is the weighted adjacency matrix (the GRN to infer), and $\boldsymbol{Z}$ is a random noise term.

Rearranging into a VAE:

$$\text{Encoder:} \quad \boldsymbol{Z} = \boldsymbol{X}(\boldsymbol{I} - \boldsymbol{A}^\top) \qquad \text{Decoder:} \quad \boldsymbol{X} = \boldsymbol{Z}(\boldsymbol{I} - \boldsymbol{A}^\top)^{-1}$$

The adjacency matrix $\boldsymbol{A}$ is a learnable parameter, jointly optimized with the encoder/decoder networks via:

$$\mathcal{L} = -\mathbb{E}_{\boldsymbol{z} \sim q(\boldsymbol{Z}|\boldsymbol{X})}[\log p(\boldsymbol{X}|\boldsymbol{Z})] + \beta \cdot D_{\mathrm{KL}(q(\boldsymbol{Z}|\boldsymbol{X}) \,\|\, p(\boldsymbol{Z}))} + \alpha \cdot \|\boldsymbol{A}\|_1$$

The ℓ_1 penalty on $\boldsymbol{A}$ encourages sparsity (real GRNs are sparse).

Key limitations:

1. **Training instability:** DeepSEM's inferred networks degrade as training continues past convergence, likely due to overfitting dropout noise.
2. **Matrix inversion:** The $(\boldsymbol{I} - \boldsymbol{A}^\top)^{-1}$ operation is numerically unstable and computationally expensive.
3. **Static reconstruction:** The model reconstructs static snapshots; it doesn't model the *dynamics* that generate gene expression.
4. **Undirected edges in practice:** While $\boldsymbol{A}$ is asymmetric in principle, the model averages $|\boldsymbol{A}|$ over multiple runs, partially losing directionality.
5. **No knowledge sharing:** Every dataset is trained from scratch—no transfer of regulatory logic across cell types, tissues, or species.

1.2 Proposed Architecture: FlowGRN

We propose replacing the VAE generative backbone with **conditional flow matching (CFM)**, while preserving the core insight of embedding the GRN structure into the generative model. Critically, the latent-space formulation enables a natural decomposition into **shared** (transferable) and **dataset-specific** (GRN) components.

1.2.1 Encoder: Per-Gene MLP with Shared Weights

Following DeepSEM's design, a small MLP processes one gene at a time with shared weights:

$$h_i = \mathrm{MLP}_{\mathrm{enc}(x_i)} \quad \text{for each gene } i \in \{1, ..., g\}$$

$$\boldsymbol{z} = \left[h_1, h_2, ..., h_g\right] \in \mathbb{R}^{c \times g \times d}$$

The shared-weight design ensures gene-gene interactions are *not* embedded in the encoder—they are exclusively captured by the adjacency structure in the velocity field. Alternatively, a **pretrained encoder** from scGPT or Geneformer can be used. The encoder weights θ_{enc} are **shared across all datasets** in the multi-task/transfer setting.

1.2.2 GRN-Structured Velocity Field (Core Innovation)

The velocity field $v_\theta(\boldsymbol{z}_t, t)$ is parameterized as a **Graph Attention Network** operating over genes. For gene i at flow time t:

$$v_i(\boldsymbol{z}_t, t) = \sum_{j=1}^{g} \alpha_{ij} \cdot \mathrm{MLP}_{\mathrm{msg}\left(z_t^j, t\right)} + \mathrm{MLP}_{\mathrm{self}(z_t^i, t)}$$

The attention weights incorporate a learnable adjacency bias:

$$e_{ij} = \mathrm{LeakyReLU}\Big(\boldsymbol{a}^\top \cdot \Big[\boldsymbol{W} z_t^i \parallel \boldsymbol{W} z_t^j \parallel \gamma(t)\Big]\Big)$$

$$\alpha_{ij} = \frac{\exp\big(e_{ij} + \lambda \cdot A_{ij}\big)}{\sum_k \exp(e_{ik} + \lambda \cdot A_{ik})}$$

where:

- $\boldsymbol{A} \in \mathbb{R}^{g \times g}$ is the **learnable adjacency matrix** (the GRN),
- λ controls the strength of the adjacency prior vs. data-driven attention,
- $\gamma(t)$ is a sinusoidal time embedding.

Key insight: The velocity at gene i depends on other genes j weighted by α_{ij}. High $|A_{ij}|$ means gene j has strong regulatory influence on gene i's dynamics.

1.2.3 Flow Matching Training

Instead of the VAE's reconstruction + KL loss, we train with the **conditional flow matching (CFM) objective**:

$$\mathcal{L}_{\mathrm{CFM}} = \mathbb{E}_{t, \boldsymbol{x}_0, \boldsymbol{x}_1} \left\| v_\theta(\boldsymbol{x}_t, t) - u_t(\boldsymbol{x}_t \mid \boldsymbol{x}_0, \boldsymbol{x}_1) \right\|^2$$

where $\boldsymbol{x}_0 \sim p_{\mathrm{noise}}$ (e.g., standard Gaussian), $\boldsymbol{x}_1 \sim p_{\mathrm{data}}$ (scRNA-seq expression vectors), $\boldsymbol{x}_t = (1-t) \cdot \boldsymbol{x}_0 + t \cdot \boldsymbol{x}_1$ (linear interpolation), and $u_t(\boldsymbol{x}_t \mid \boldsymbol{x}_0, \boldsymbol{x}_1) = \boldsymbol{x}_1 - \boldsymbol{x}_0$ (target velocity for the linear path).

1.2.4 Full Loss Function

$$\mathcal{L} = \mathcal{L}_{\mathrm{CFM}} + \alpha \cdot \|\boldsymbol{A}\|_1 + \beta \cdot \mathcal{L}_{\mathrm{DAG}}$$

where $\mathcal{L}_{\mathrm{DAG}} = \mathrm{tr}\big(e^{\boldsymbol{A} \circ \boldsymbol{A}}\big) - g$ is an optional acyclicity constraint (from NOTEARS) that penalizes cyclic structures, and $\circ$ denotes the Hadamard product.

1.2.5 GRN Extraction

After training: (1) $|A_{ij}|$ gives regulatory strength of gene $j \to$ gene i; (2) $\text{sign}(A_{ij})$ indicates activation ($+$) or repression ($-$); (3) rank all gene pairs by $|A_{ij}|$ for evaluation.

1.3 Variant: Latent Flow Matching

For better noise handling, operate the flow in a compressed latent space:

1. **Pretrain an autoencoder** on scRNA-seq data: Encoder $\mathcal{E} : \mathbb{R}^g \to \mathbb{R}^d$ ($d \ll g$), Decoder $\mathcal{D} : \mathbb{R}^d \to \mathbb{R}^g$.
2. **Run flow matching in latent space:** $\boldsymbol{z}_0 \sim \mathcal{N}(0, \boldsymbol{I}_d)$, $\boldsymbol{z}_1 = \mathcal{E}(\boldsymbol{x})$, train velocity field on $\boldsymbol{z}_t$ trajectories.
3. **Extract GRN from velocity field Jacobian:** Compute $\frac{\partial v_i}{\partial z_j}$ at representative points, then map back to gene space via the decoder Jacobian $\frac{\partial \boldsymbol{x}}{\partial \boldsymbol{z}}$.

1.4 Extension: Transfer Learning via Shared-Private Decomposition

The latent-space formulation enables a natural decomposition into **shared** (transferable) and **private** (dataset-specific) components. Current GRN methods (DeepSEM, DAZZLE, GENIE3, GRNBoost2) all train from scratch on each dataset independently.

1.4.1 Biological Rationale

Gene regulation has two distinct layers:

1. **Conserved regulatory grammar**—the *mechanism* by which TFs bind DNA, how signaling cascades propagate, how chromatin remodeling works. Broadly shared across cell types and partially conserved across species.
2. **Context-specific regulatory wiring**—*which* TFs are active, *which* enhancers are accessible, *which* feedback loops are engaged. These differ between hESC, mDC, and hHEP.

FlowGRN's architecture naturally separates these:

Component	What it captures	Shared / Private
Encoder (MLP_{enc})	Gene expression $\to$ latent mapping	**Shared**
Velocity MLPs (MLP_{msg}, MLP_{self})	How regulatory signals propagate	**Shared**
Attention parameters ($\boldsymbol{W}_Q$, $\boldsymbol{W}_K$, $\boldsymbol{a}$)	How to compute influence strength	**Shared**
Adjacency matrix $\boldsymbol{A}_k$	Which edges are active in cell type k	**Private**
Dataset embedding $\boldsymbol{e}_k$ (optional)	Cell-type-specific context	**Private**

Table 1: Shared-private decomposition of FlowGRN parameters.

1.4.2 Transfer Variants

Variant T1: Joint Multi-Task Training. Train one FlowGRN on all K datasets simultaneously:

$$\mathcal{L}_{\text{total}} = \sum_{k=1}^{K} w_k \cdot \mathcal{L}_{\text{CFM}}^{(k)}(\theta_{\text{shared}}, \boldsymbol{A}_k, \boldsymbol{e}_k) + \alpha \cdot \sum_k \|\boldsymbol{A}_k\|_1$$

where $\theta_{\text{shared}} = \{\theta_{\text{enc}}, \theta_{\text{MLP_msg}}, \theta_{\text{MLP_self}}, \boldsymbol{W}_Q, \boldsymbol{W}_K, \boldsymbol{a}\}$ is trained on all datasets, and each $\boldsymbol{A}_k$ is trained only on dataset k.

Variant T2: Pretrain-then-Finetune. Stage 1: train jointly on K datasets $\rightarrow$ get θ^*_{shared}. Stage 2: for a new target dataset, freeze θ^*_{shared} and only train $\boldsymbol{A}_{\text{target}}$:

$$\boldsymbol{A}^*_{\text{target}} = \arg\min_{\boldsymbol{A}_{\text{target}}} \mathcal{L}^{\text{target}}\big(\theta^*_{\text{shared}}, \boldsymbol{A}_{\text{target}}\big) + \alpha \cdot \|\boldsymbol{A}_{\text{target}}\|_1$$

Variant T3: Cross-Species Transfer. For organisms with ortholog mappings (e.g., mouse $\leftrightarrow$ human), introduce an ortholog matrix $\boldsymbol{O} \in \{0,1\}^{g_h \times g_m}$ where $O_{ij} = 1$ if human gene i is orthologous to mouse gene j. Warm-start:

$$\boldsymbol{A}_{\text{human}} \leftarrow \boldsymbol{O} \cdot \boldsymbol{A}_{\text{mouse}} \cdot \boldsymbol{O}^\top$$

then finetune $\boldsymbol{A}_{\text{human}}$ on human data with θ_{shared} frozen.

Variant T4: Foundation Model Backbone. Use a frozen pretrained encoder from scGPT (33M cells) or Geneformer (30M cells). Only train the velocity field (shared MLPs + per-dataset $\boldsymbol{A}_k$).

1.4.3 Dataset Conditioning

Optionally condition the velocity field on dataset identity:

$$v_i(\boldsymbol{z}_t, t; \boldsymbol{A}_k, \boldsymbol{e}_k) = \sum_j \alpha_{ij}^{(k)} \cdot \text{MLP}_{\text{msg}\left([\boldsymbol{z}_t^j; \gamma(t); \boldsymbol{e}_k]\right)} + \text{MLP}_{\text{self}([\boldsymbol{z}_t^i; \gamma(t); \boldsymbol{e}_k])}$$

where $\boldsymbol{e}_k \in \mathbb{R}^{d_e}$ is a learnable embedding for dataset k, and $[;]$ denotes concatenation.

1.5 Architecture Comparison

Feature	DeepSEM/DAZ-ZLE	GRNFormer	FlowGRN	FlowGRN + Transfer
Generative model	VAE	N/A (discrim.)	Flow Matching	Flow Matching
GRN param.	Adj. in enc/dec	Graph transf.	Adj-biased GAT	Per-dataset $\boldsymbol{A}_k$
Training obj.	Recon. + KL	Supervised	CFM (unsup.)	Multi-task CFM
Dynamics	No (static)	No	Yes (vel. field)	Yes
Directionality	Weak ($\lvert\boldsymbol{A}\rvert$ avg)	TF-subgraphs	Strong	Strong
Cross-dataset	None	None	None	**Yes**
Few-shot GRN	No	Needs labels	No	**Yes**
Cross-species	No	No	No	**Yes**

Table 2: Comparison of FlowGRN with existing GRN inference methods.

2 Mathematical Formulation

2.1 Conditional Flow Matching: Foundations

A **continuous normalizing flow (CNF)** defines a time-dependent velocity field $v_\theta : \mathbb{R}^d \times [0,1] \to \mathbb{R}^d$ generating a flow ψ_t via the ODE:

$$\frac{d\psi_t(\boldsymbol{x})}{dt} = v_\theta(\psi_t(\boldsymbol{x}), t), \qquad \psi_0(\boldsymbol{x}) = \boldsymbol{x}$$

The flow transports a source distribution p_0 to a target distribution p_1: $p_t = (\psi_t)_{\#} p_0$.

Flow Matching Objective (Lipman et al., 2022):

$$\mathcal{L}_{\text{FM}} = \mathbb{E}_{t\sim\mathcal{U}[0,1],\, \boldsymbol{x}\sim p_t} \, \|v_\theta(\boldsymbol{x}, t) - u_t(\boldsymbol{x})\|^2$$

This is intractable since p_t and u_t are unknown. The key insight is to condition on endpoints.

Conditional Flow Matching (CFM): Define conditional probability paths $p_t(\boldsymbol{x} \mid \boldsymbol{z})$ with known conditional velocity fields $u_t(\boldsymbol{x} \mid \boldsymbol{z})$ such that $p_t(\boldsymbol{x}) = \int p_t(\boldsymbol{x} \mid \boldsymbol{z}) q(\boldsymbol{z}) d\boldsymbol{z}$. Then:

$$\mathcal{L}_{\text{CFM}} = \mathbb{E}_{t\sim\mathcal{U}[0,1],\, \boldsymbol{z}\sim q(\boldsymbol{z}),\, \boldsymbol{x}\sim p_t(\cdot \mid \boldsymbol{z})} \, \|v_\theta(\boldsymbol{x}, t) - u_t(\boldsymbol{x} \mid \boldsymbol{z})\|^2$$

has the same gradients as $\mathcal{L}_{\text{FM}}$ with respect to θ.

2.2 OT-CFM with Minibatch Couplings

Standard CFM uses independent coupling: $\boldsymbol{z} = (\boldsymbol{x}_0, \boldsymbol{x}_1)$ sampled independently. This leads to crossing paths and higher training variance.

OT-CFM uses an optimal transport coupling:

$$\pi^* = \arg\min_{\pi \in \Pi(p_0, p_1)} \int \|\boldsymbol{x}_0 - \boldsymbol{x}_1\|^2 \, d\pi(\boldsymbol{x}_0, \boldsymbol{x}_1)$$

In practice, **minibatch OT** approximates this. Given a batch of B noise samples $\{\boldsymbol{x}_0^i\}_{i=1}^B$ and B data samples $\left\{\boldsymbol{x}_1^j\right\}_{j=1}^B$, solve the assignment:

$$\sigma^* = \arg\min_{\sigma \in S_B} \sum_{i=1}^{B} \|\boldsymbol{x}_0^i - \boldsymbol{x}_1^{\sigma(i)}\|^2$$

via the Hungarian algorithm ($\mathcal{O}(B^3)$) or Sinkhorn approximation.

The conditional paths for OT-CFM are:

$$\boldsymbol{x}_t = (1-t) \cdot \boldsymbol{x}_0 + t \cdot \boldsymbol{x}_1 \qquad u_t(\boldsymbol{x} \mid \boldsymbol{x}_0, \boldsymbol{x}_1) = \boldsymbol{x}_1 - \boldsymbol{x}_0$$

Why this matters for GRN inference: OT coupling produces straighter, non-crossing flows. Since gene expression distributions have complex multimodal structure (different cell types), OT coupling ensures the velocity field captures biologically meaningful transitions rather than arbitrary noise-to-data mappings.

2.3 Applying CFM to GRN Inference

Setup for single-cell data:

- Source distribution p_0: Standard Gaussian $\mathcal{N}(0, \boldsymbol{I}_g)$

- Target distribution p_1: Empirical distribution of scRNA-seq expression vectors
- Each sample $\boldsymbol{x} \in \mathbb{R}^g$ is a cell's expression profile across g genes

GRN-structured velocity field:

$$v_{\theta,i}(\boldsymbol{x}_t, t) = \sum_{j \in \mathcal{N}(i)} \alpha_{ij}(\boldsymbol{x}_t, t) \cdot \varphi\Big(x_t^j, t\Big) + \psi(x_t^i, t)$$

where $\mathcal{N}(i)$ is the neighborhood of gene i, α_{ij} are attention weights with adjacency bias, and φ, ψ are shared MLPs.

Attention mechanism with adjacency bias:

$$e_{ij} = \mathrm{LeakyReLU}\Big(\boldsymbol{a}^\top \cdot \Big[\boldsymbol{W}_Q \boldsymbol{x}_t^i \parallel \boldsymbol{W}_K \boldsymbol{x}_t^j \parallel \gamma(t)\Big]\Big)$$

$$\alpha_{ij} = \frac{\exp\big(e_{ij} + \lambda \cdot A_{ij}\big)}{\sum_k \exp(e_{ik} + \lambda \cdot A_{ik})}$$

Full training loss (single-dataset):

$$\mathcal{L} = \mathbb{E}_{t \sim \mathcal{U}[0,1],\, (\boldsymbol{x}_0, \boldsymbol{x}_1) \sim \pi_{\mathrm{OT}}} \|v_\theta(\boldsymbol{x}_t, t) - (\boldsymbol{x}_1 - \boldsymbol{x}_0)\|^2 + \alpha \cdot \|\boldsymbol{A}\|_1 + \beta \cdot \mathcal{R}(\boldsymbol{A})$$

where $\mathcal{R}(\boldsymbol{A})$ is an optional structural regularizer:
- **Acyclicity (NOTEARS):** $\mathcal{R}(\boldsymbol{A}) = \mathrm{tr}\big(e^{\boldsymbol{A} \circ \boldsymbol{A}}\big) - g$
- **Degree constraint:** $\mathcal{R}(\boldsymbol{A}) = \sum_i \max\Big(0, \sum_j |A_{ij}| - \kappa\Big)$
- **Prior incorporation:** $\mathcal{R}(\boldsymbol{A}) = \|\boldsymbol{A} \circ \boldsymbol{M}\|_F^2$ where $\boldsymbol{M}$ is a mask from TF motif data

2.4 Multi-Task Flow Matching for Transfer Learning

Given K datasets $\{\mathcal{D}_1, ..., \mathcal{D}_K\}$, decompose parameters into:
- $\theta_{\text{shared}} = \big\{\theta_{\text{enc}}, \theta_{\text{MLP_msg}}, \theta_{\text{MLP_self}}, \boldsymbol{W}_Q, \boldsymbol{W}_K, \boldsymbol{a}, b_A, \lambda\big\}$
- $\theta_{\text{private}} = \{\boldsymbol{A}_1, ..., \boldsymbol{A}_K, \boldsymbol{e}_1, ..., \boldsymbol{e}_K\}$

Multi-task loss:

$$\mathcal{L}_{\text{total}} = \sum_{k=1}^{K} w_k \cdot \mathcal{L}^{(k)}(\theta_{\text{shared}}, \boldsymbol{A}_k, \boldsymbol{e}_k) + \alpha \cdot \sum_k \|\boldsymbol{A}_k\|_1 + \beta \cdot \sum_k \mathcal{R}(\boldsymbol{A}_k)$$

where:

$$\mathcal{L}^{(k)} = \mathbb{E}_{t,\, \left(\boldsymbol{x}_0, \boldsymbol{x}_1^{(k)}\right) \sim \pi_{\mathrm{OT}}^{(k)}} \|v_{\theta_{\text{shared}}}(\boldsymbol{x}_t, t; \boldsymbol{A}_k, \boldsymbol{e}_k) - \Big(\boldsymbol{x}_1^{(k)} - \boldsymbol{x}_0\Big)\|^2$$

2.4.1 Gradient Flow Analysis

Gradients from $\mathcal{L}^{(k)}$ flow into θ_{shared} but are blocked from other datasets' adjacency matrices:

$$\frac{\partial \mathcal{L}^{(k)}}{\partial \theta_{\text{shared}}} \neq 0, \qquad \frac{\partial \mathcal{L}^{(k)}}{\partial \boldsymbol{A}_k} \neq 0, \qquad \frac{\partial \mathcal{L}^{(k)}}{\partial \boldsymbol{A}_j} = 0 \quad (j \neq k)$$

This ensures each $\boldsymbol{A}_k$ reflects only dataset k's GRN, while θ_{shared} captures cross-dataset regulatory patterns.

2.4.2 Transfer to New Datasets

For a new dataset $\mathcal{D}_{\text{target}}$:

$$\boldsymbol{A}^*_{\text{target}} = \arg\min_{\boldsymbol{A}_{\text{target}}} \mathcal{L}^{\text{target}}\big(\theta^*_{\text{shared}}, \boldsymbol{A}_{\text{target}}, \boldsymbol{e}_{\text{target}}\big) + \alpha \cdot \|\boldsymbol{A}_{\text{target}}\|_1$$

with θ^*_{shared} frozen from pretraining.

2.4.3 Cross-Species Formulation

Introduce ortholog mapping $\boldsymbol{O} \in \{0,1\}^{g_h \times g_m}$ ($O_{ij} = 1$ iff human gene i is orthologous to mouse gene j). Warm-start the human adjacency:

$$\boldsymbol{A}^{\text{init}}_{\text{human}} = \boldsymbol{O} \cdot \boldsymbol{A}_{\text{mouse}} \cdot \boldsymbol{O}^\top$$

then finetune $\boldsymbol{A}_{\text{human}}$ on human data with θ_{shared} frozen.

2.5 GRN Extraction and Scoring

Method 1: Direct adjacency. Edge score$(j \to i) = |A_{ij}|$; edge sign$(j \to i) = \text{sign}\big(A_{ij}\big)$.

Method 2: Jacobian analysis. For representative cells $\boldsymbol{x}^*$ (cluster centroids):

$$J_{ij}(\boldsymbol{x}^*, t) = \frac{\partial v_i(\boldsymbol{x}^*, t)}{\partial x_j}$$

$$\text{GRN}_{ij} = \mathbb{E}_{t,\boldsymbol{x}^*}\big[\!\big[J_{ij}(\boldsymbol{x}^*, t)\big]\!\big]$$

Method 3: Integrated Gradients.

$$\text{IG}_{ij} = \int_0^1 \frac{\partial v_i(\boldsymbol{x}_t, t)}{\partial x_j} \cdot \left(x_1^j - x_0^j\right) dt$$

Transfer-specific note: In the multi-task setting, Methods 2 and 3 capture effects from both $\boldsymbol{A}_k$ (dataset-specific) and θ_{shared} (universal dynamics), potentially yielding more informative GRNs than the raw $\boldsymbol{A}_k$ alone.

2.6 Biological Interpretation of the Flow

- $v_i(\boldsymbol{x}, t) > 0$: gene i is being upregulated at state $\boldsymbol{x}$, flow time t
- α_{ij} large: gene j strongly influences gene i's rate of change $\to$ regulatory edge
- **Flow trajectories:** integrating v_θ from noise to data traces cell state transitions
- **Cell-type-specific GRNs:** evaluate $J_{ij}(\boldsymbol{x}^*, t)$ at different cell-type centroids
- **Shared vs specific regulation:** edges appearing in all $\boldsymbol{A}_k$ are conserved; edges unique to one $\boldsymbol{A}_k$ are context-specific

3 Experimental Setup

3.1 BEELINE Benchmark: Datasets and Ground Truths

A) Synthetic Networks (6 datasets). Generated via BoolODE: dyn-LI (linear), dyn-CY (cyclic), dyn-LL (long linear), dyn-BF (bifurcating), dyn-TF (trifurcating), dyn-BFC (bifurcating converging). Each with $\{100, 200, 500, 2000, 5000\}$ cells $\times$ 10 samples. Ground truth: exact generating network.

B) Curated Boolean Models (4 datasets). HSC (hematopoietic stem cell), mCAD (cortical area development), VSC (ventral spinal cord), GSD (gonadal sex determination). 2000 cells $\times$ 10 samples, plus 50% and 70% dropout variants. Ground truth: Boolean model structure.

C) Experimental scRNA-seq (7 datasets). hESC (human embryonic stem cells), hHEP (human hepatocytes), mDC (mouse dendritic cells), mESC (mouse embryonic stem cells), mHSC-E/GM/L (mouse hematopoietic stem cells, three lineages). Ground truths: non-specific ChIP-seq, STRING, cell-type-specific ChIP-seq, and LOF/GOF networks.

3.2 Evaluation Metrics

1. **AUPRC ratio:** AUPRC divided by baseline (where baseline = edge density). Primary metric.
2. **EPR** (Early Precision Ratio): precision among top K predicted edges (K = # ground truth edges).
3. **AUROC**: secondary metric; less informative for sparse networks.

3.3 Baselines

Tier 1 (DL, unsupervised, on BEELINE): DeepSEM, DAZZLE (current SOTA), GRN-VAE, HyperG-VAE.

Tier 2 (Classical): GENIE3, GRNBoost2, PIDC, PPCOR.

Tier 3 (Recent SOTA): GRNFormer, scKAN, scRegNet, GRANGER.

Tier 4 (Transfer-aware): GRNPT, Meta-TGLink, scMTNI, LINGER.

3.4 Experimental Protocol

3.4.1 Experiment 1: BEELINE Benchmark—Single-Dataset

All three BEELINE categories. TFs + 500 and TFs + 1000 genes. 10 random seeds per dataset. Key comparisons: FlowGRN vs DAZZLE; OT coupling vs independent; with/without DAG constraint.

Hyperparameter search: $d \in \{32, 64, 128\}$; $\alpha \in \{0.001, 0.01, 0.1\}$; $\lambda \in \{0.1, 1.0, 10.0\}$; batch OT size $B \in \{64, 128, 256\}$; GAT heads $\in \{1, 4, 8\}$; flow steps $\in \{10, 20, 50, 100\}$.

3.4.2 Experiment 2: Ablation Studies

Velocity field architecture (GAT vs MLP vs Transformer); OT coupling type; GRN extraction method (direct $\boldsymbol{A}$ vs Jacobian vs IG); latent vs gene space; dropout robustness (0%, 50%, 70%); with/without dropout augmentation from DAZZLE.

3.4.3 Experiment 3: External Biological Validation

A) ChIP-seq validation: Predict TF-target edges, validate against ENCODE ChIP-seq. Report AUC/AUPR per TF.

B) Perturbation prediction: Use inferred GRN to predict TF knockout effects against Perturb-seq and CRISPRi screens.

C) Cross-species transfer: Train on well-annotated species, evaluate on related species.

3.4.4 Experiment 4: Scalability Analysis

Runtime and memory vs number of genes (100–10,000) and cells (100–20,000). Compare against DeepSEM, DAZZLE, GENIE3, GRNBoost2.

3.4.5 Experiment 5: Multi-Task Joint Training

Test whether sharing regulatory dynamics across BEELINE datasets helps.

- **FlowGRN-Solo:** trained independently on each dataset (baseline)
- **FlowGRN-Joint:** trained on all 7 datasets (shared θ, per-dataset $\boldsymbol{A}_k$)
- **FlowGRN-Joint-Cond:** same as Joint with dataset conditioning embeddings $\boldsymbol{e}_k$

Report per-dataset AUPRC/EPR/AUROC. Analyze which datasets benefit most—hypothesis: smaller datasets and datasets from the same organism/lineage benefit most.

3.4.6 Experiment 6: Few-Shot GRN Inference

Leave-one-out: For each dataset k, pretrain on the other 6 $\rightarrow$ get θ^*_{shared}, then finetune only $\boldsymbol{A}_k$ (frozen θ_{shared}). Compare against Solo from scratch.

Data titration: Subsample to $\{100\%, 50\%, 20\%, 10\%, 5\%\}$ of cells. Compare Solo vs Transfer at each level. Plot AUPRC vs # cells.

Expected: transfer helps most in low-data regime. If pretrained FlowGRN with 10% cells matches Solo with 100%, this is a strong result for rare cell types.

3.4.7 Experiment 7: Cross-Species Transfer (Mouse $\rightarrow$ Human)

1. **Source:** Train jointly on 5 mouse datasets (mDC, mESC, mHSC-E/GM/L) $\rightarrow \theta^*_{\text{shared, mouse}}$
2. **Ortholog mapping:** Ensembl BioMart one-to-one orthologs ($\sim$16,000 genes)
3. **Target:** Transfer to human datasets (hESC, hHEP):
   - Warm start: $\boldsymbol{A}_{\text{human}} \leftarrow \boldsymbol{O} \cdot \boldsymbol{A}_{\text{mouse,avg}} \cdot \boldsymbol{O}^\top$
   - Cold start: $\boldsymbol{A}_{\text{human}} = \boldsymbol{0}$
   - Finetune $\boldsymbol{A}_{\text{human}}$ with θ_{shared} frozen
4. **Controls:** Reverse direction (human $\rightarrow$ mouse); same-species transfer

3.4.8 Experiment 8: Shared vs Specific Regulation Analysis

A) Conserved edges: Compare $\boldsymbol{A}_k$ across datasets. Edges with high $|A_{ij}|$ in all datasets $\rightarrow$ conserved regulation. Validate against known housekeeping interactions.

B) Context-specific edges: High in one $\boldsymbol{A}_k$, low in others $\rightarrow$ context-specific wiring. Validate against cell-type-specific ChIP-seq.

C) Shared weight inspection: Analyze what MLP_{msg} learns universally. Visualize attention before/after adjacency bias.

D) Latent space structure: UMAP of shared latent space across datasets. Do cells undergoing similar processes cluster across datasets?

3.5 Implementation Plan

Libraries: PyTorch, TorchCFM, PyTorch Geometric, Scanpy, POT, pybiomart.

Hardware: Single GPU (A100/H100); models trainable in minutes per dataset.

```
flowgrn/
├── models/
│   ├── encoder.py, velocity_field.py, flow_matching.py
│   ├── grn_extraction.py, transfer.py
├── data/
│   ├── beeline_loader.py, preprocessing.py
│   ├── multitask_loader.py, ortholog_mapper.py
├── evaluation/
│   ├── metrics.py, beeline_eval.py, transfer_analysis.py
└── experiments/
    ├── exp1–exp8 scripts
```

3.6 Expected Outcomes and Risk Mitigation

Core FlowGRN: 5–15% AUPRC improvement over DAZZLE; lower variance across seeds; better on high-dropout data; stronger directionality; competitive runtime.

Transfer extension: Joint training $\geq$ Solo on most datasets; pretrained FlowGRN at 20% data $\approx$ Solo at 100%; cross-species transfer outperforms Solo when human data is limited; biologically interpretable conserved vs specific edges.

Risk	Mitigation
Flow matching $<$ VAE on static snapshots	Add temporal datasets; use pseudotime for source$\rightarrow$target
OT computation too expensive	Sinkhorn with $\varepsilon = 0.1$; top-k genes per batch
Adjacency doesn't converge	Initialize $\boldsymbol{A}$ from GENIE3 (warm start); anneal α
Marginal BEELINE gains	Focus on bio validation + transfer novelty
Negative transfer in joint training	Report honestly; Solo baseline always included
Cross-species transfer fails	Same-species transfer as control to isolate species barrier
Low gene overlap between datasets	Focus on TF-centric subnetwork where overlap is higher
Shared latent space collapses	Monitor UMAP/silhouette; add auxiliary reconstruction loss

Table 3: Risk mitigation strategy.

4 Novelty Claims and Related Work

1. **vs TrajectoryNet/TIGON/CellOT:** These use OT/flow for *trajectory inference* with GRN as a downstream byproduct. FlowGRN embeds the GRN *directly into the velocity field.*
2. **vs CycleGRN:** Restricted to oscillatory/cell-cycle genes. FlowGRN is general-purpose.
3. **vs DeepSEM/DAZZLE:** Replaces the VAE with flow matching (no matrix inversion, OT-based noise handling, interpretable velocity field). The shared-private architecture enables transfer learning impossible in the SEM framework.
4. **vs GRNFormer:** Supervised, requires labels. FlowGRN is fully unsupervised.

5. **vs scGPT/Geneformer for GRN:** Foundation models extract embeddings for post-hoc GRN inference. FlowGRN learns the GRN as a structural component of the generative model—but can *use* foundation encoders as its backbone (Variant T4).
6. **vs GRNPT:** Uses LLM text embeddings + supervised training. FlowGRN achieves transfer via shared generative dynamics, entirely unsupervised.
7. **vs Meta-TGLink:** Supervised meta-learning for few-shot GRN. FlowGRN achieves few-shot via pretrained generative dynamics without ground-truth labels.
8. **vs scMTNI:** Shares information via multi-task regularization on single lineages. FlowGRN shares through learned generative dynamics, applies to any dataset collection.
9. **vs LINGER:** Uses external bulk data via lifelong learning. FlowGRN transfers from other single-cell datasets—complementary approaches that could be combined.

Target venues: Nature Methods, Nature Computational Science, Genome Biology (full scope); Bioinformatics (core method); NeurIPS/ICML (ML angle with bio validation).